

Conservation of Momentum for 2D Collisions Activity

10
10

Purpose:

To investigate if momentum is conserved during a 2D collision.

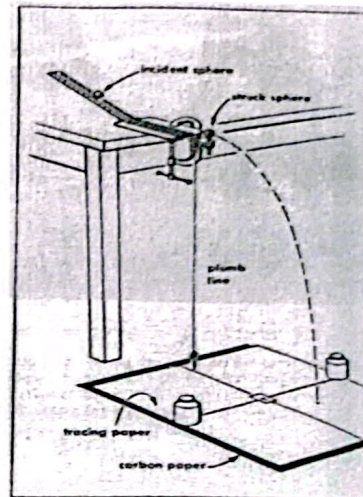
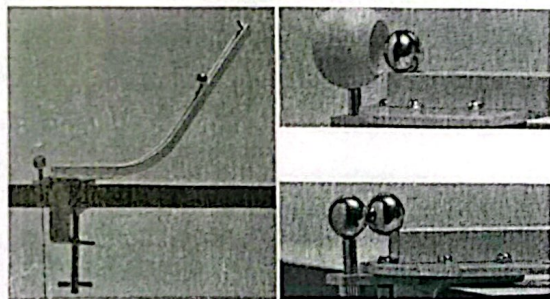
Materials:

2D collision apparatus, two steel balls, clamp, plumb line and bob, carbon paper, masking tape and meter stick

Experimental Set Up:

The basic setup resembles the diagram. Some special notes about the setup:

- The flat base plate must extend over the table so the plumb line touches the floor
- Ramp must be directly aligned with the target post; balls must contact at same height
- Alignment is correct when one sound is heard and both balls land at the same time
- Four pieces of regular white paper must be taped to the floor to make a base
- Four pieces of carbon paper are taped to cover the regular paper

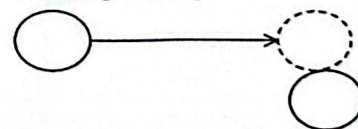


*****NOTE LAYING DOWN THE PAPERS IS THE LAST STEP WHEN ALL ADJUSTMENTS AND TRIALS ARE COMPLETE**

Procedure:

1. Determine the initial horizontal momentum of the incident ball before the collision, by releasing it 7 times. Circle the distribution of 'landings' on the paper. Circle these marks and label the centre as P_{original}
2. Place an identical steel ball on the target support. Swivel the support sideways about 45 degrees to produce a glancing collision. Look right for "birds eye view."

Glancing side tap



3. The two balls should hit the paper at two different points. Repeat 7 times to create a cluster of landings. Draw a circle around these and mark the centres as T_{after} for the ball that was stationary (target), and P_{after} for the other (projectile).

Observations:

4. Draw vector lines from the mark at the plumb lines to the centre of each of the three groups of dots created.
5. Measure and record the length of the displacement vectors and angles in degrees by filling in the data table.

Generate your own data table in the space provided. You need to measure the displacement vectors and angles before and after the collisions . . .

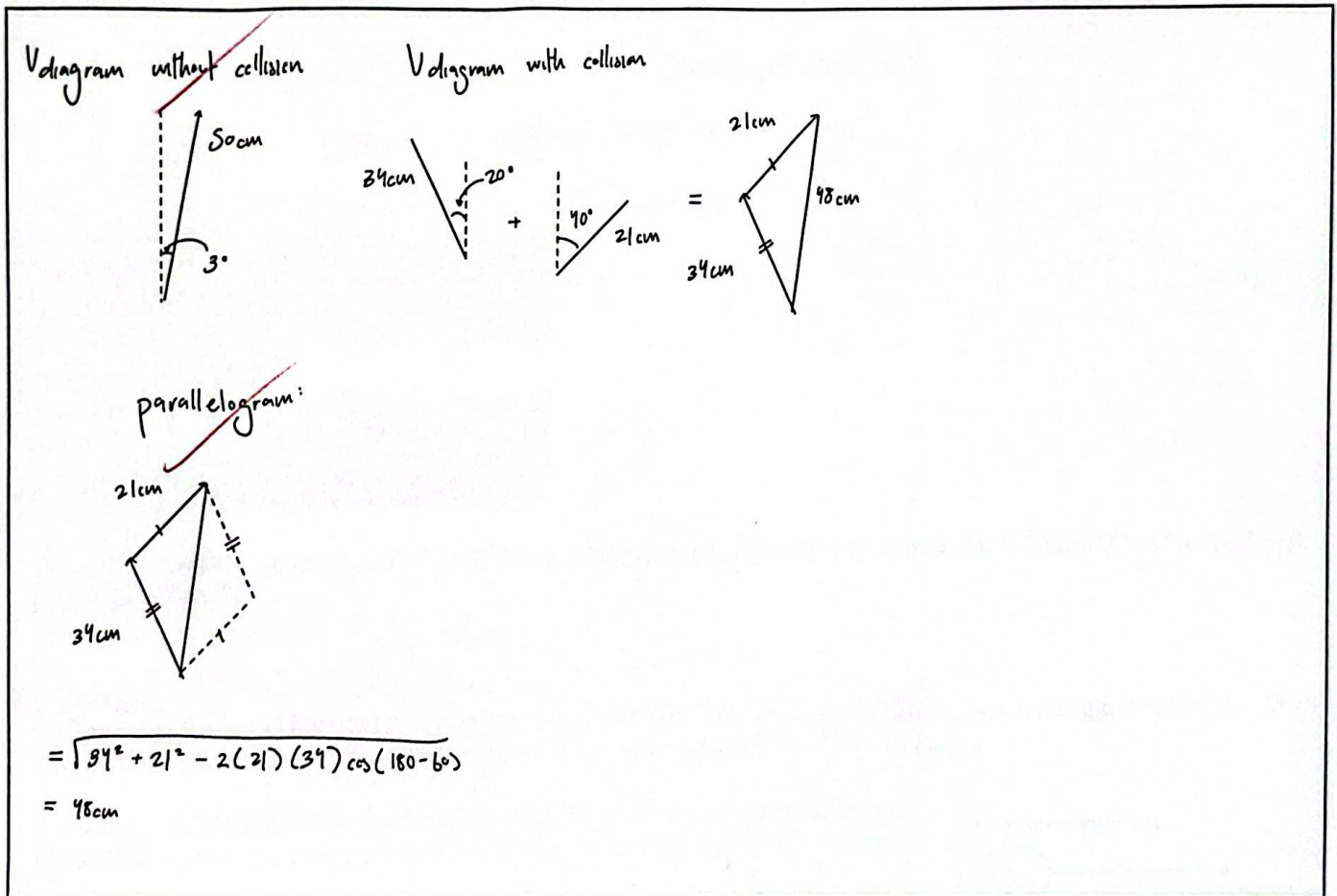
data / observation table

Crystal be

Original	Patter	Patter
50cm [N3°E]	21cm [N40°E]	34cm [N20°W]

Analysis:

6. Draw a scaled vector diagram that will show if the momentum vectors after the collision are equal to the momentum vector before the collision. Complete the parallelogram to compare the vector sum of the collision pairs with the original momentum vector. *Note, the displacement vectors are actually proportional to the momentum vectors.*



Conclusion:

Notice the angle between the two momentum vectors after the collision. Does this angle have any significance?

Yes, the angle between the 2 momentum vectors does have significance. It tells us if the momentum was conserved or distributed through the collision. A larger angle signifies the momentum was spread more evenly on the 2 balls, while a smaller angle indicates greater momentum on one ball, rather than both.

Does your evidence indicate that momentum was conserved in these collisions? What errors may have been introduced into this experiment? How might these errors be reduced?

The evidence suggests that momentum was largely conserved. The vector sum of the collision vectors is very similar to the original, without collision vector. This highlights how momentum is largely conserved. However, the sum of the collision vectors is slightly shorter, at 48cm compared to the original 50cm. This may be because of a multitude of reasons such as the balls experiencing air resistance and friction which would have caused a loss of momentum. In order to mitigate such discrepancies, it could be run in a controlled environment, like a vacuum. Other possible sources of error are human errors, such as measurement inaccuracies, which would lead to inaccurate vectors and calculations. Other errors may include variation in initial height. To remedy these errors, one could use a mechanism that outputs uniform releases. Additionally, the material of the balls may have caused them to absorb energy. To mitigate this, use balls made of an